

Algorithmic Codesign of Federated Distillation

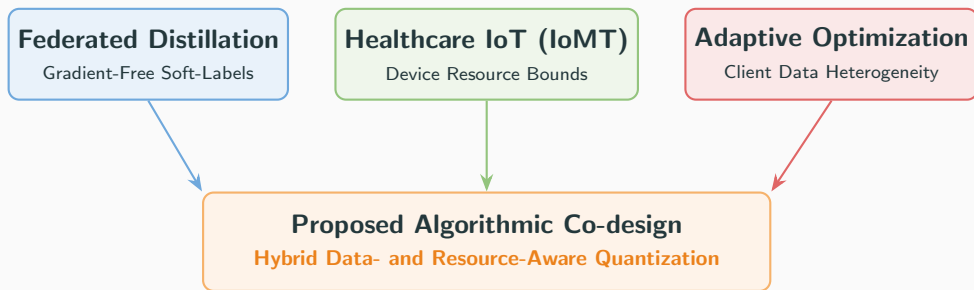
Research Progress Update

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Research Domains



Convergence of soft-label communication, local hardware constraints, and data distribution complexity.

The Compression-Utility Trade-off

- Static compression baseline methods destroy fine-grained activity recognition features.
- Hardware-only dynamic triggers blindly over-compress critical motion data in skewed settings.
- We aim to balance communication savings with patient fall-recognition accuracy.

Literature Gaps

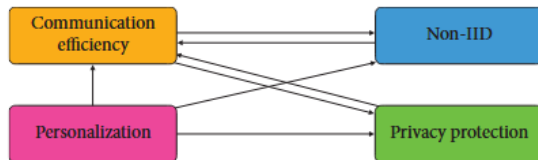


FIGURE 9: The relationships between different challenges in FL. Increased communication helps alleviate non-IID but may lead to privacy concerns. Using too many privacy-preserving technologies can lead to a communication burden. Mitigating non-IID helps reduce communication. Personalization helps alleviate communication bottlenecks, privacy concerns, and non-IID.

Source: Qin et al. (2025), Figure 9: Relationships between challenges in FL.

[Placeholder]

Cue: Open NotebookLM mind maps.

[https://notebooklm.google.com/notebook/
96030b06-c526-477e-8282-d3ca7c50ee92?authuser=1](https://notebooklm.google.com/notebook/96030b06-c526-477e-8282-d3ca7c50ee92?authuser=1)

Edge System Constraints

- Smart wearable sensors suffer from strict energy boundaries and battery exhaustion.
- High peak memory overhead limits active local training loops directly on edge devices.
- Suboptimal fixed scheduling policies fail under highly variable and heterogeneous networks.

Edge Architecture Hierarchy

[Placeholder: Intelligent Edge Systems]

Cue: Open Cajas Ordóñez et al. (2025), Figure 6 ("Architectures and algorithms for resource management on a fog/edge computing model") as it is too detailed to fit inline.

Clinical Personalization

- Cloud-trained models suffer from reduced generalization across individual patient biometrics.
- Personalized transfer learning adapts general models to user-specific wearable sensor signals.
- Federated transfer loops ensure patient health data remains local and highly secure.

FedHealth Personalization Loop

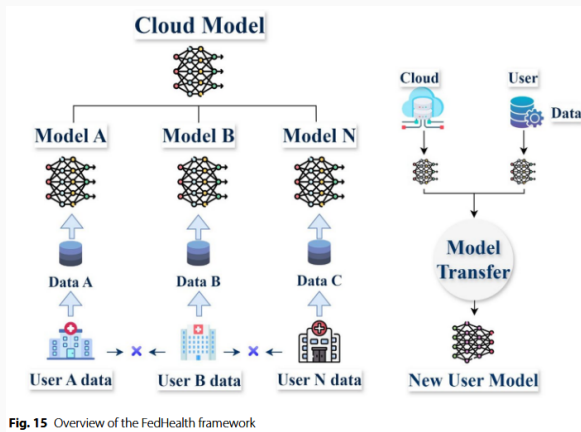


Fig. 15 Overview of the FedHealth framework

Source: Salman et al. (2025), Figure 15: Overview of the FedHealth framework.

Model Compression

Compression Gaps

- Traditional quantization and pruning techniques significantly reduce client-side overhead.
- Over-compression destroys life-critical local movement features on low-resource wearables.
- Active dynamic feedback loops are necessary to adapt to statistical client data variance.

Compression Classification

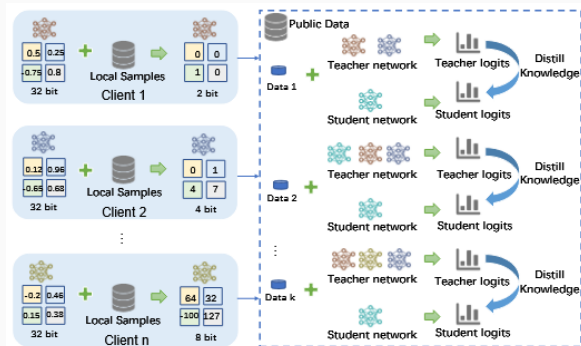


Figure 2: An overview of the training procedure proposed by DynFed.

Source: He et al. (2025), Figure 2: Overview of the training procedure proposed by DynFed.

Federated Distillation

Distillation Mechanics

- Client nodes exchange soft-label logits rather than raw network weights.
- Transmission sizes on communication channels are drastically minimized.
- Logit aggregation naturally mitigates label skew under non-IID conditions.

Knowledge Distillation Flow

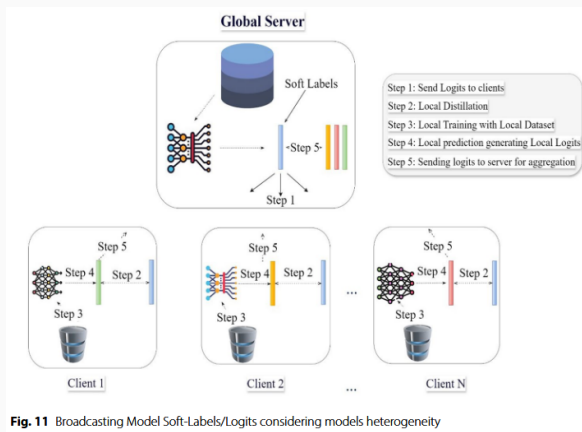


Fig. 11 Broadcasting Model Soft-Labels/Logits considering models heterogeneity

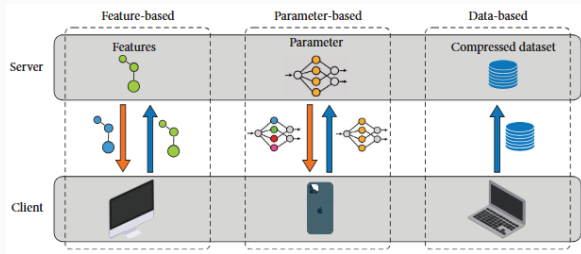
Source: Salman et al. (2025), Figure 11: Broadcasting model soft-labels/logits considering model heterogeneity.

Paradigm Comparison

Federated Distillation Taxonomy

- Feature-based FD shares only model outputs, significantly reducing communication overhead.
- Parameter-based FD uploads student networks, addressing model fusion constraints.
- Data-based FD exchanges synthetic datasets, lowering local compute costs.

KD-based Federated Learning Comparison



Source: Qin et al. (2025), Figure 5: Comparison of the three KD-based FL methods.

Workspace Organization



BSMSCS Thesis Hub

A hub of repositories for my BS/MS Computer Science thesis. It will be continuously updated until I graduate.

 Philippines  <https://www.notion.so/cjbnyi/BSMSCS...>  [in/christian-joseph-bunyi](https://www.linkedin.com/in/christian-joseph-bunyi)  christian_bunyi@dlsu.edu.ph

README .md 

Hi there 🙋

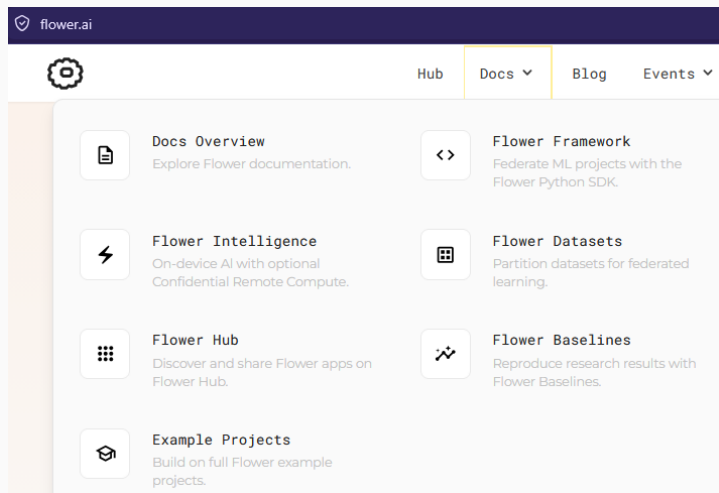
This is a work in progress.

Pinned Customize pins

 [reproducibility-artifacts](#) Public

 Python

Simulation Framework

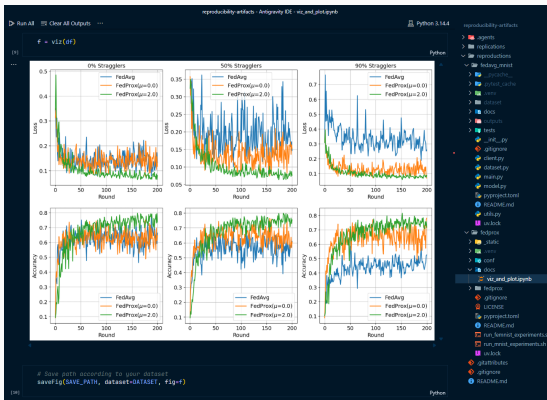


Baseline Replication

Local Baseline Execution

- Replicated classic FedAvg and FedProx optimization baselines successfully on local hardware.
- Configured standard SGD optimizers on wearable sensor datasets.
- Validated training and inference pipelines under varying local epoch configurations.

Validation Curves



Thank You!

Questions & Answers